

George Ortiz

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EDUCATION

Carnegie Mellon University, M.S. in Mechanical Engineering - Research, GPA: 4.0

May 2027

Stevens Institute of Technology, B.E. Mechanical Engineering, Robotics Concentration, GPA: 3.75

Grad. May 2025

RESEARCH

Robotics Researcher, Robomechanics Lab

Aug 2025 – Present

- Research collaboration with FieldAI to train a learned controller for adaptive quadrupedal locomotion through non-Newtonian fluids, with handling of up to 20% penetration in deep mud.
- Trained policies on a Bingham plastic model of non-Newtonian forces for training in IsaacLab, and developed a validation sim-to-sim step in NVIDIA Newton and Genesis-world, with two-way force coupling between Newton's MPM solver and the robot's rigid body solver.
- Ran hardware testing for projects involving traversing across a balance beam and compliant entanglements, tuning and debugging estimation and calibration.
- Integrated MuJoCo physics backend into Quad-SDK (previously Gazebo only), wrote a state estimator to bridge MuJoCo sensor outputs to RoS2, enabling contact-rich simulation capability not possible in the original stack.

Robotics Engineer, PROOF Lab

Feb 2023 – May 2025

- Developed object detection algorithm for Doosan H2515 collaborative robot with YOLOv8 and OpenCV in Python, classifying and locating objects to enhance robot task execution.
- Parameterized space and developed a mapping algorithm to automate pick-and-place tasks with millimeter precision, making functions for path planning and accelerating robotics research.
- Created Python-based client-server architecture using sockets in an IoT setup, emulating TCP communication which improved reliability and simplified robot-computer interaction.

WORK EXPERIENCE

Manufacturing Engineer Intern, Zimmer Biomet Holdings

May - Aug 2024

- Created a fixture in Siemens NX using GD&T to maximize safety after identifying a manual operation hazard.
- Investigated and managed large datasets in scrap data and designed a dashboard in Power BI, enabling management to analyze scrap data with real-time analysis and reduced manual reporting time by 40%.
- Recognized challenges with fixture setup at laser etch operation, made programs for over 100-part families in DELMIA, embedded visual aids which increased clarity, and reduced training time.

Quality Engineer Intern, Zimmer Biomet Holdings

Sep - Dec 2023

- Conducted time studies and Quality Standard Reference (QSR) updates for an inspection reduction project, resulting in \$40,000 in annual cost savings.
- Owned and completed an A3 project for cases containing excessive errors made by operators, checking to ensure error rate was within a reasonable margin.

ROBOTICS PROJECTS

Soft Exosuit for patients with Spinal Muscular Atrophy (SESMA)

ME Capstone Project

- Addressed muscle weakness caused by Spinal Muscular Atrophy, improving mobility for affected users by 10%.
- Programmed microcontroller and integrated PID compensation in C++ to monitor sit-to-stand transitions for real-time feedback and control.
- Collected and analyzed data in MATLAB with 3D splines for curve fitting to calculate changes in cable length, optimizing clutch release for maximum knee torque assistance.

Signature SO-100 Robot Manipulator

LeRobot NYC Arm Hackathon

- Separated mechanical and electrical noise of potentiometer outputs using leader and follower paths and ran low pass filter through the noise to smooth trajectory, accelerating tasks with a 200% speed increase.
- Trained a robot arm to autonomously pick up a pen from anywhere within the workspace, draw on a piece of paper, and place a pen down through an Action Chunking Transformer (ACT) Policy.
- Won \$1000 prize for "Most Innovative Use of Data".

SKILLS

Software Isaac Sim/Lab & Newton Branch, MuJoCo, RoS2, YOLOv8, OpenCV, PyTorch

Languages Python, C++, MATLAB, Git, Linux

Algorithms Model Free Reinforcement Learning, Model Predictive Control, SLAM, RRT, Extended Kalman Filters